Lab 6 Technical Memo

E80

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Abstract

This experiment measured a variety of objects in a wind tunnel then simulated these objects with computational fluid dynamics (CFD) software in order to understand the effects on a drag given the object being measured as well as the efficacy of the simulation and its ability to represent a prototype. This will be achieved with an adjustable wind tunnel, the "Theory of Models", COMSOL software, and analysis of drag forces on objects [1]. Incidentally, this memo finds it practical to utilize the Theory of Models in order to represent prototypes and there is evidence the shape of an object moving through a fluid (as represented in simulations and practice) has effect on drag force but not the drag coefficient of an object as long as surface area remains unchanged.

Introduction

Conducting this analysis serves great importance in the understanding of fluid dynamics and mechanical design when considering movement through fluids (i.e. air, water). The concepts of fluid dynamics can be represented both through simulation, and practical applications. However, they may differ when conditions are not ideal which is the reason this experiment utilized the Theory of Models. This theory claims that a prototype of an object can be sufficiently represented by a model if they obey the same set of parameters (or are both "described by a set of Π terms") [1]. In this case, these parameters would be Reynold's number compared against both the drag force and drag coefficient of each object. This can help judge the performance of the model and the performance of each of the objects.

The evaluation of both the model and the different objects depends on a set of parameters as outlined above: drag force, drag coefficient, and Reynold's number. The experiment primarily focused on drag which is the resistance of the fluid on an object acting opposite to its motion. Drag force can be denoted by the equation:

$$F_D=\frac{1}{2}\rho v^2 C_D A$$

where ρ is the density of the fluid, v is the velocity of the object, C_D is the drag coefficient, and A is the cross sectional area. Drag is important in judging the performance of each of the objects as it directly responds to the shape of the object moving through the fluid. One of its arguments, the drag coefficient, is "a number that aerodynamicists use to model all of the complex dependencies of drag on shape, inclination, and some flow conditions" [2]. This number is a constant in any given object and a direct result of the shape of said object and how a fluid would react/resist. Since it is constant and dimensionless, the Theory of Models should be applicable and its value can be examined to determine the performance of a CFD model. The drag coefficient, C_D, of any given object can be found by the following equation:

$$C_D=\frac{2Fd}{\rho v^2 A}$$

This is a rearrangement of the previous equation where the choice of reference area affects the way an object reacts. This number is integral and contains "not only the complex dependencies of object shape and inclination, but also the effects of air viscosity and compressibility" [2]. In the equation, Fd is the drag force and the rest of the parameters obey what was said before.

Finally and most importantly, Reynolds number (Re) is the ratio of inertial forces to viscous forces and is yet another dimensionless parameter [1]. In essence, this number determines the type of flow experienced and the way the fluid will react given the system characteristics. This experiment looks primarily at laminar flow (which is a smoother kind of flow in fluid dynamics) and will have Reynold's number be an independent variable. By controlling Reynold's number and the system, the model and prototype can each examine how drag and drag coefficients respond. Reynold's number can be calculated with the following equation:

$$Re = \frac{pvL}{\mu}$$

Here, ρ is the fluid density, v remain upstream velocity, L is characteristic length of the container, and μ is the fluid viscosity. The two dimensionless parameters, Re and C_D, will serve to examine the efficacy of the model and how it compares against three different prototypes. These prototypes are elaborated below.

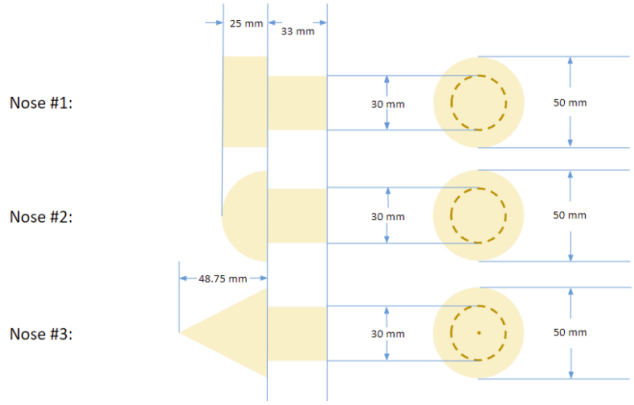


Figure 1: The 3 different nose cones being observed, each shares the same cylindrical base and is segmented with the extrusion on these nose cones

This memo also serves to describe mechanical design and why the shape of an object can contribute significantly to the amount of drag that object can feel. The objects of interest in terms of drag force, drag coefficient, velocity profile, and Reynold's number were three different kinds of cones, hereby referred to as Nose #1, #2, and #3. As Figure 1 depicts, Nose #1 is cylindrical and flat faced, Nose #2 is a hemisphere, and Nose #3 is conical. Each is situated on the same cylinder that has a radius of 25 mm and height of 125 mm. This experiment expects that both the model and prototype will show a proportional increase in drag force as Reynold's number increases. Nose #3 should experience the least drag out of the three and the drag coefficient should remain constant for each different cone regardless of the value of Reynold's number.

Methods

In order to obtain measurements across different nose cone shapes, the experiment made use of a wind tunnel which provided measured lift and drag values. As NASA describes, a wind tunnel serves as a controlled environment where parameters like velocity, density, area and more are adjustable so this means Reynold's number can also be controlled [2]. The wind tunnel serves to provide test data to aerodynamicists and examine the forces on an aircraft given certain flow conditions [3]. The wind tunnel has a fan which can be set at a certain percentage of max fan speed. This value was translated/calibrated into upstream velocity in order to analyze the data and directly compare it to the computational model. The data was measured at particular intervals of percentage of max fan speed. There are also drag values at zero flow velocity that must be subtracted from measurements since they are instrument offsets.

In the prototype scenario of this experiment, a cylindrical object (and an associated nose cone shape) is mounted in a chamber, and the resulting lift and drag values could be shown on the LVDT box at any controlled flow velocity. These intervals are also controlled in the simulation with laminar flow and other similar considerations in mind (as to preserve the Theory of Models).

Once again, the fluid dynamics simulations made use of the COMSOL software, where different nose cones were constructed then put inside a constructed "test chamber" box where a fluid dynamics simulation would be performed. The cones were set as acrylic plastic and the box it was encased in was set as air, and such by creating an upstream velocity of the fluid, the air drag across each cone can be modeled computationally. COMSOL features a "fluid flow interface [that] defines sets of domain equations, boundary conditions, initial conditions, predefined meshes, predefined studies with solver settings for steady and transient analyses, as well as predefined plots and derived values" [4]. By setting the objects and fluids apart, defining laminar flow, then inputting the desired parameters (which are replicated from the prototype), the shear stress in a given direction can be calculated by the program. The shear stress acting parallel to the direction of the flow would also be known as the drag force. By changing parameters, the computational model can continuously calculate drag force (and the associated coefficient) as Reynold's number changes.

Results

The first notable result is the efficacy of COMSOL's CFD interface and whether or not the simulated steady and transient analyses are sufficient to represent the prototype tested in the wind tunnel. Referring back to the background of the module, there are expectations that each test must fulfill in order to be a good representation of the principles of fluid dynamics. First, the drag coefficient is meant to show the drag or resistance of an object against a fluid. Thus, whether simulated or measured, this value should remain constant and independent of Reynold's number.

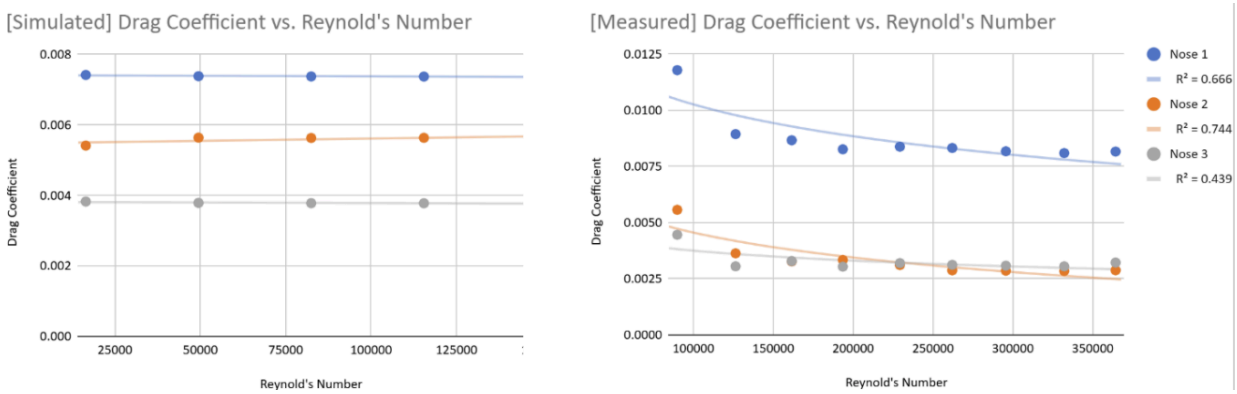


Figure 2: Simulated and Measured plots of drag coefficients against Reynold's number, shown for all 3 nose cone shapes. Lines of best fit have been shown for each set of data points

The CFD interface used in this experiment seems to be consistent with expectations. As Figure 2 depicts on the left, Drag coefficient is relatively flat and consistent regardless of the parameters changed in the software. Thus, it makes sense this value is a property of the object being measured. Each Nose has a different value for drag coefficient and that is likely a result of their geometry and how it is expected to react against a fluid. Nose #3, in gray, has the lowest drag coefficient which indicates it experiences less aerodynamic drag and experienced less resistance overall when in the "test chamber" box. Another interesting thing to note about Figure 2 is on the right. The measured drag coefficient comes from the prototypes tested in the wind tunnel. Across all three noses, there is a discrepancy in the first drag coefficient value recorded. This could be a result of the wind tunnel used throughout the course of the

analysis and how that data point was recorded. It's likely the wind tunnel was not completely controlled and fluid density and/or viscosity varied throughout the chamber. Despite that discrepancy, the drag coefficient slopes do lie around a similar threshold as their simulated counterparts. However, the measured drag coefficient of Nose #2 does seem oddly close to Nose #3 and quite far from its simulated counterpart. It's likely this was an experimental error or a further consequence of the non-ideal conditions of the wind tunnel.

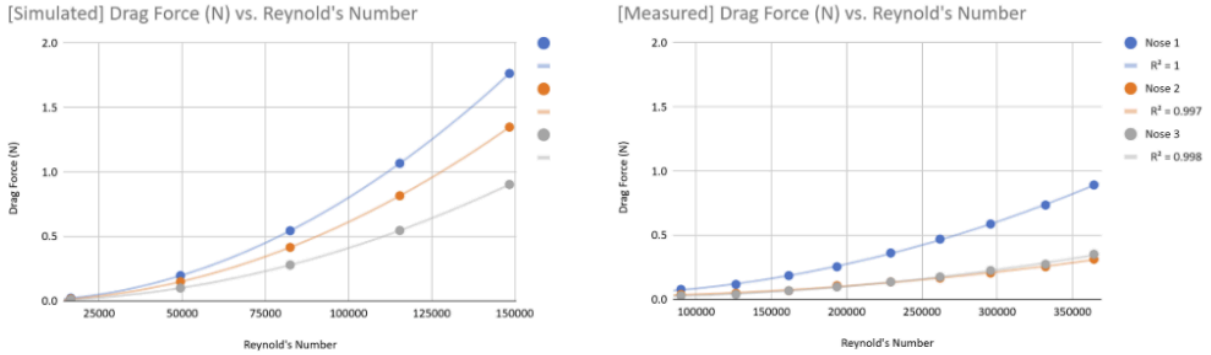


Figure 3: Two plots outlining the difference in nose cones and how they experience drag and the difference between measurements and simulations

As expected, there are deviations between the simulated and measured results in this general experiment, which could be attributed to a variety of factors. Figure 3 demonstrates that the measured values of drag force did not increase as significantly as when simulated. The errors may be a result of unpredictability and the way the air in the wind tunnel interacted with the object. A particular parameter may be at fault for the scale. For example, the pressure, humidity and/or density of the air are variable around the chamber. Another explanation is that at high Reynolds numbers, the type of flow may change as thus, the laminar flow simulations may not be appropriate for the prototype. This is because the simulation It's also interesting to see the measured results show that same similarity between Nose #2 and Nose #3. This similarity is likely an error made in measurement given the concreteness of the theory and model. Drag is a force parallel to the direction of flow and thus the similarities in the nose cone of #2 and #3 are likely a testament to the similarities in the measured results [5].

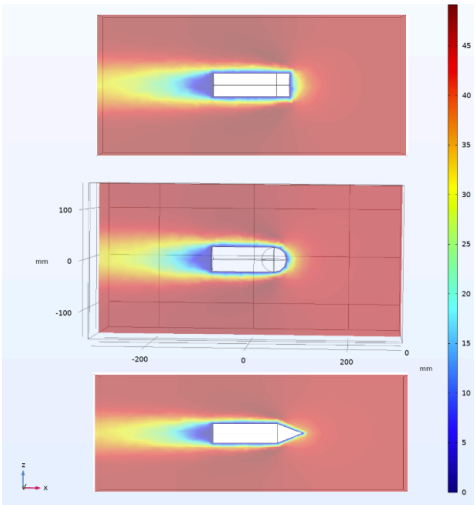


Figure 4: A velocity profile of all three nose cones in the x- and z-axis provided by the COMSOL software. The gradient legend on the right represents velocity in m/s.

Figure 4 shows the velocity profile of each nose cone and their cylindrical base and how the velocity of the fluid (flowing to the left in each case) varies when interacting with a given object. Along with a pressure contour diagram, this profile is generated by the calculations made in COMSOL as is part of the CFD interface. This particular comparison is significant because it experimentally shows a concept of fluid dynamics and what the given shape of an object can do when interacting with a fluid. Figure 4 demonstrates Nose #3 having the least resistance in its head. Correspondingly, it has the least amount of blue space or lower velocity in a given area than the other two. The aerodynamic qualities of Nose #3 give it an edge and disperse the drag forces on the object, allowing for less disturbance in velocity of the fluid.

Conclusion

The measurement of these types of nose cones and their performance in fluids (via simulation and experimental measurements) has applications in oceanography and serves as a good basis for mechanical design of vehicles moving through fluids. Fluid dynamics is a fundamental consideration of the mechanical design of underwater robotics as to reduce drag on our object. By obeying the principles of aerodynamics, fluid dynamics, and utilizing the Theory of Models, a reliable prototype and final design can be constructed. The Theory of Models, in particular, can aid in the engineering design such that simulation and a smaller-scale prototype can represent a larger system.

Despite the discrepancies, it appears the COMSOL modeling performed in this analysis was a reasonable and accurate way of evaluating the different nose cones. The results were as expected and the principles of fluid dynamics were observed and examined sufficiently across both processes. Nose #3, which did feature the most “aerodynamic” design, did perform better than the other two models by reducing drag and dispersing force in an effective manner. Drag force, drag coefficient, and Reynold's number also proved to be a good basis for assessing both the model’s efficacy and the performance of each nose cone. Methods of analysis were also developed throughout the course of the experiment in order to accurately represent the results and present them throughout this memo. There are many other complex intricacies of fluid dynamics that could’ve factored into the data of this experiment. Further experiments could examine other kinds of flow or a varying chamber with properties dependent on temperature.

References

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[2] T. Benson, “The Drag Coefficient,” www.grc.nasa.gov. https://www.grc.nasa.gov/www/k-12/VirtualAero/BottleRocket/airplane/dragco.html
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